Mississippi Valley State University

CS 341 — Discrete Structures

Practice Set 12: Approved Solutions

Union, Intersection, Difference, and Complement

Covers Chartrand & Zhang, Section 2.2, pp. 60–68

Status of this set

Ungraded practice. Solutions released with it. Attempt everything first.

Part I: How to Practice This Material

Translate to logic, then compute

The dictionary

$x \in A \cap B$	$(x \in A) \wedge (x \in B)$
$x \in A \cup B$	$(x \in A) \vee (x \in B)$ — *inclusive*
$x \in \overline{A}$	$\sim(x \in A)$
$x \in A - B$	$(x \in A) \wedge \sim(x \in B)$
$x \in A \oplus B$	$(x \in A) \oplus (x \in B)$

Any question about sets can be turned into a question about statements, and you have four weeks of practice with those.

Three mistakes to expect

1. **Reading $\cup$ as exclusive.** An element in both sets is still in the union.
2. **Assuming $A - B = B - A$.** There is no commutative law for difference.
3. **Forgetting the universal set.** $\overline{A}$ is meaningless until U is named.

Part II: Practice Problems

Section A — Computing

1. Let $U = \{1, 2, \ldots, 10\}$, $A = \{1, 2, 3, 4, 5\}$, $B = \{4, 5, 6, 7\}$, and $C = \{2, 4, 6, 8\}$. Compute each.

(a) $A \cap B$

(b) $A \cup B$

(c) $A - B$

(d) $B - A$

(e) $A \oplus B$

(f) $\overline{A}$

(g) $\overline{A \cup B}$

(h) $A \cap B \cap C$

Approved solution

(a) $\{4, 5\}$ **(b)** $\{1, 2, 3, 4, 5, 6, 7\}$ **(c)** $\{1, 2, 3\}$ **(d)** $\{6, 7\}$
(e) $A \oplus B = (A - B) \cup (B - A) = \{1, 2, 3, 6, 7\}$ — everything in exactly one of the two.
(f) $\overline{A} = \{6, 7, 8, 9, 10\}$ **(g)** $\overline{A \cup B} = \{8, 9, 10\}$
(h) $A \cap B \cap C = \{4\}$ — only 4 is in all three.

Check (g) against De Morgan: $\overline{A} \cap \overline{B} = \{6, 7, 8, 9, 10\} \cap \{1, 2, 3, 8, 9, 10\} = \{8, 9, 10\}$. Same answer, as Theorem 2.35 requires. Doing that check costs nothing and catches arithmetic slips.

2. With the same U, A, B as above, verify by direct computation that $A \cap B \subseteq A \cup B$, and explain why this holds for *every* pair of sets.

Approved solution

$A \cap B = \{4, 5\}$ and $A \cup B = \{1, \ldots, 7\}$, and every element of the first is in the second.
In general: if $x \in A \cap B$ then $x \in A$ and $x \in B$; in particular $x \in A$, so x is in at least one of the two and therefore in $A \cup B$.
Logically this is just $(P \wedge Q) \Rightarrow (P \vee Q)$, which is a tautology — you classified exactly this as Homework 4, Problem 1(a). **And it is the reason $\cup$ must be the inclusive or:** with an exclusive reading, elements of $A \cap B$ would be excluded from the union and the containment would fail.

Section B — Membership tables

3. Build a membership table for $A - B$ and for $A \cap \overline{B}$, and state whether they are equal.

$x \in A$	$x \in B$	$x \in \overline{B}$	$x \in A - B$	$x \in A \cap \overline{B}$
T	T	F	F	F
T	F	T	T	T
F	T	F	F	F
F	F	T	F	F

Approved solution

The last two columns agree in all four rows, so $A - B = A \cap \overline{B}$.
This is a genuinely useful identity: it lets you eliminate the difference operator entirely and work with only $\cap$, $\cup$, and complement. Both sides say the same thing — $(x \in A) \wedge \sim(x \in B)$.

Section C — Reasoning

4. Answer each with a brief justification.

(a) If $A \subseteq B$, what is $A \cap B$? What is $A \cup B$?

(b) If A and B are disjoint, what is $A \oplus B$?

(c) Can $A - B = B - A$ ever hold? Under what condition?

Approved solution

(a) $A \cap B = A$ and $A \cup B = B$. Every element of A is already in B, so the overlap is all of A and the union adds nothing to B.
(b) $A \oplus B = A \cup B$. With no overlap to remove, "in exactly one" and "in at least one" coincide. Compare Practice Set 2, Problem 8(b): when P and Q cannot both be true, $\vee$ and $\oplus$ agree. Disjointness is that condition for sets.
(c) Yes — exactly when $A = B$, in which case both differences are $\emptyset$.
If $A \neq B$, some element lies in one and not the other; say $x \in A$ and $x \notin B$. Then $x \in A - B$ but $x \notin B - A$, so the two differ. Hence equality forces $A = B$.

Section D — Writing

5. In five to eight complete sentences, explain why a Venn diagram is useful for *understanding* a set identity but does not *prove* it. Say what does constitute a proof, and mention what a membership table covers that a picture does not.

Approved solution

Model answer. A Venn diagram shows one particular arrangement of the sets, drawn so that the regions overlap in a generic way, and shading it makes the claim of an identity easy to see. But a picture is a single illustration: it does not establish that the claim holds for every possible collection of sets, and a reader who doubts the identity has been shown a drawing rather than an argument. A membership table does better, because its rows enumerate every way an element can sit relative to the sets — four rows for two sets, eight for three — and checking that two columns agree in all rows leaves no case untested. A double-inclusion argument does better still, since it takes an arbitrary element and reasons about it without assuming how many sets there are or how they overlap. So the diagram is a tool for seeing what to prove, and the table or the element argument is the proof itself.

Full credit requires the point that a diagram is *one case* while a table is *all cases*. An answer saying only "pictures aren't rigorous" earns about half.

Section E — Challenge (optional)

6. Two sets carve the universe into four regions, and three sets into eight. How many regions do n sets create, and what does that have to do with membership tables?

Approved solution

2^n regions. Each set independently either contains a given element or does not, so an element's position is described by n independent yes-or-no choices.
That is exactly the row count of a membership table on n sets — because a row *is* a region. Verifying an identity by table is the same as checking every region of the Venn diagram, which is why the table counts as a proof and a single drawing does not.

And it is 2^n again: binary passwords in Lecture 1, truth-table rows in Week 1, SAT's blow-up in Week 3, power sets in Week 4, and now Venn regions. Same counting argument, fifth context.
Practical note: this is why Venn diagrams are rarely drawn for more than three sets. Four sets need sixteen regions, which cannot be produced with circles at all.

Instructor notes

- **Problem 2 is the one to put on the board.** $A \cap B \subseteq A \cup B$ holds *because* union is inclusive. It retroactively justifies a Week 1 definition students found arbitrary at the time.
- **Problem 3** gives $A - B = A \cap \overline{B}$, which students will use constantly in Homework 6. Worth naming as a result rather than an exercise.
- **Problem 4(b)** connects disjointness to the Week 1 fact that $\vee$ and $\oplus$ agree when both cannot hold. Same observation, new setting.
- **Problem 6** is the fifth appearance of 2^n this term. Students notice the pattern by now; naming it explicitly makes the course feel like one subject.
- Reserved for Homework 6: even-numbered exercises from Section 2.2. Odd ones have printed answers at pp. 596–597 and are self-check only.